Supplementary Material for Hippocampal Controllability across Clinically Defined Alzheimer's Disease-Related Groups: Associations with Structural Network Topology and Group Discrimination

1 Supplementary Methods

Throughout the Supplementary Material, BH-FDR-adjusted p values are denoted by p_{FDR}, whereas unadjusted and permutation p values are denoted by p and p_{perm}, respectively.

1.1 Cortical and Structural-Connectivity Visualizations

For descriptive cortical visualization, ROI-level average and modal controllability values were averaged within each clinical group and mapped to the Schaefer2018 cortical surface using a common color scale (Supplementary Fig. S1). Group-averaged structural connectivity matrices were also used to visualize hippocampal connections and connections between Tian atlas-defined non-cortical structures and the seven Schaefer networks. Only relatively high-weight connections were retained for display; this visualization threshold was not used in any inferential analysis. Mean connectivity between each atlas-defined structure group and cortical network was summarized in heatmaps (Supplementary Fig. S2).

1.2 Classification Analyses

The four binary tasks were CN versus AD, SMC versus MCI, MCI versus AD, and CN versus SMC. Six low-dimensional feature sets were examined: global, left, and right hippocampal controllability; bilateral FreeSurfer hippocampal volume; hippocampal strength and betweenness ; and their six-feature combination. Controllability predictors were within-subject ranks, volume and topology were unranked, and the combined model contained both scales. The 24 task–feature combinations were exploratory and formed one Holm-corrected family. All models used inverse-frequency-balanced L_2-regularized logistic regression and the same covariates: age, sex, and education; ICV was not a classification covariate. Ten repetitions of five-fold stratified outer cross-validation generated held-out probabilities, while four-fold stratified inner cross-validation selected $C \in \{0.01, 0.1, 1, 10\}$. Covariate residualization and standardization were fitted within each inner-training partition during hyperparameter selection and refitted on the corresponding outer-training data before held-out prediction. Each participant's 10 held-out probabilities were averaged arithmetically before calculation of one subject-level AUC.

Classifier significance used 1,000 complete-pipeline label permutations for every task–feature combination. Labels were permuted before cross-validation; every permutation reran the full repeated nested procedure, including foldwise residualization, standardization, inner selection of C, model refitting, and outer prediction. Permutation p values used the plus-one correction. AUC confidence intervals used 5,000 group-stratified bootstrap resamples of the fixed subject-level mean out-of-fold probabilities. These intervals quantify subject-sampling uncertainty conditional on the fitted predictions and do not fully capture training-set variability; the mean, standard deviation, and range of the 10 repeat-specific AUCs are reported in Table II. Direct candidate-versus-volume comparisons reran both models in the same complete-case sample using identical splits and the same random seed as Table II. Paired AUC differences used 5,000 group-stratified bootstrap resamples and subject-level score-swap tests; the 20 comparisons formed a separate Holm-corrected family (Supplementary

Table S16). The high-dimensional connectivity analysis remains exploratory; its previous fixed-score permutation values are removed and only descriptive AUC estimates are retained.

1.3 Primary Rank Endpoints and Unranked-Value Sensitivity Analyses

The primary endpoints were mean within-subject ranks: nodal average and modal controllability were ranked separately from 1 to 450 within each participant and then averaged across the bilateral hippocampal parcels. Adjusted group means, 95% confidence intervals, and partial η^2 were obtained from the full group-plus-demographics models using 5,000 group-stratified bootstrap resamples. Six pairwise contrasts per endpoint were tested with two-sided Freedman–Lane permutations and controlled jointly as one 12-test BH-FDR family.

Unranked bilateral hippocampal values were analyzed after adjustment for age, sex, education, and $\log(1 + W_{\text{total}})$, where $W_{\text{total}} = \sum_{i<j} A_{ij}$ and A was the final inverse-node-volume-scaled connectivity matrix before spectral stabilization. A further model added connectome density. Unranked omnibus and density-adjusted omnibus tests formed separate two-outcome BH-FDR families, and the 12 unranked pairwise contrasts formed another family. Direct average–modal associations used partial Spearman correlations: rank-based measures were adjusted for age, sex, and education, whereas unranked measures were additionally adjusted for matrix scale.

1.4 Cohort Construction and Diagnostic Sensitivity Analysis

The cohort was reconstructed from the ADNI imaging-metadata export used for the analysis, the final subject manifest, and the executable audit code. A session was a unique participant-date combination. Same-date duplicate diffusion series were resolved using the lower numeric image identifier, and the earliest eligible date was retained when multiple dates remained. The previous 240 participants were retained; systematic application of the same eligibility criteria identified 9 additional SMC and 32 additional AD participants. The archived materials supported a reconstructed query inventory and partial cohort-selection audit. They did not preserve the historical intermediate exclusion reasons; no retrospective reason was assigned without an archived record (Supplementary Table S17).

Clinical-group assignment in the primary analysis followed the source-cohort labels for all 281 participants. Diagnostic records temporally matched to the selected diffusion-MRI sessions were examined in sensitivity analyses, which did not alter the interpretation of the primary group effects.

1.5 Cognitive Associations and Education Moderation

Partial Spearman associations between the rank-based bilateral hippocampal average- and modal-controllability measures and MMSE, ADAS-Cog 11, ADAS-Cog 13, CDR memory, and MoCA were estimated after residualizing cognition and controllability for age, sex, and education. For direct comparison with hippocampal volume, the analysis was repeated in the same FreeSurfer-complete participants after additional adjustment for ICV. BH-FDR correction was applied within each defined association family. Education was retained as a primary covariate because it differed among groups, but its potential role as a cognitive-reserve proxy was examined in separate diagnosis-by-education interaction models. The three interaction terms were tested jointly using 10,000 Freedman–Lane permutations.

1.6 Vascular and White-Matter-Hyperintensity Analyses

Modified Hachinski Ischemic Scale and UCD white-matter-hyperintensity (WMH) measurements were matched to the selected diffusion MRI session within 365 days. WMH burden was calculated as $100 \times$ TOTAL_WMH/CEREBRUM_TCV and transformed using $\log(1 + x)$. Group differences were assessed using Kruskal–Wallis tests. Partial Spearman associations with hippocampal controllability, topology, WMH burden, and cognition controlled for clinical group, age, sex, and education; BH-FDR correction was applied separately within each vascular predictor and outcome family.

1.7 MCI Subtype Audit

The ADNI DXSUM record nearest to the selected diffusion MRI session was used to audit MCI subtype. The primary window was 180 days, with 365-day and any-record audits as sensitivity checks. Memory-feature MCI was defined from DXMDES; a stricter aMCI definition additionally required DXMPTR5=Yes, indicating objective memory impairment for age and education. The primary clinical-group models were repeated after restricting the MCI group to these definitions. These classifications were used only for sensitivity analysis and do not establish subsequent conversion to AD dementia.

1.8 Amyloid PET Sensitivity Analysis

The selected diffusion MRI session was matched to the nearest visually quality-control-passed ADNI UC Berkeley 6-mm amyloid PET record within 365 days. Binary amyloid status and continuous Centiloid burden were analyzed in separate complete-case models because their availability differed; valid negative Centiloid values were retained. Models included diagnosis, age, sex, education, and either amyloid variable. Diagnosis effects adjusted for amyloid and amyloid effects adjusted for diagnosis were evaluated using 10,000 Freedman–Lane permutations. The analysis was repeated with a 730-day window. Quality-control-passed flortaucipir PET coverage was audited, but the four-group tau analysis was not interpreted because only one SMC participant had an eligible scan within 365 days.

1.9 Hippocampal Long-Axis Analysis

The Tian S3 atlas contains medial-head, lateral-head, body, and tail parcels in each hemisphere. Homologous parcels were averaged to obtain four bilateral long-axis measures from within-subject ranked controllability values. The primary clinical-group model included age, sex, and education as covariates and used 10,000 Freedman–Lane permutations. BH-FDR correction was applied across the four subdivisions separately for average and modal controllability. These macroscopic connectivity-based parcels are not equivalent to cytoarchitectonic hippocampal subfields.

1.10 Shared and Incremental Information

The source was the ADNI UCSF Cross-Sectional FreeSurfer 7.x export dated 11 July 2026. ICV, left hippocampal volume, and right hippocampal volume were ST10CV, ST29SV, and ST88SV. Eligible records were 3.0 T, had EXAMDATE exactly equal to the selected diffusion-MRI date, and contained all three measurements; same-day duplicates were resolved using the lower numeric IMAGEUID. All 270 included records used FreeSurfer 7.2.0. STATUS was partial, and OVERALLQC and HIPPOQC were not populated; volume-containing analyses were therefore exploratory.

Shared-information models predicted unranked average and modal controllability from age, sex, education, ICV, left and right hippocampal volumes entered as separate predictors, strength, and betweenness. Incremental cognitive models used the same base predictors and jointly added rank-based average and modal controllability . Reduced and full models used identical complete cases. The five two-degree-of-freedom partial F-tests formed one BH-FDR family. Percentile 95% confidence intervals for changes in R^2 used 5,000 group-stratified bootstrap resamples, with reduced and full models refitted in every resample. Because ΔR^2 is constrained to be nonnegative in nested ordinary least-squares models, these intervals were interpreted descriptively; inferential conclusions were based on the partial F-tests, with BH-FDR correction applied across the five outcomes. Clinical-group membership was analyzed using multinomial logistic regression , with a 5,000-resample percentile interval for the change in McFadden R^2.

1.11 Multiplicity Families

Multiplicity was controlled within defined analysis families rather than across every analysis in the study. The family definitions, number of tests, and correction procedures are summarized in Supplementary Table S19. Unless otherwise specified, adjusted values use BH-FDR; the exploratory classification families use Holm familywise-error correction.

1.12 Global Gray-Matter Sensitivity Analysis

Selected diffusion MRI sessions were matched to same-day ADNI UCSF Cross-Sectional FreeSurfer 7.x records. Non-hippocampal gray matter was calculated by subtracting bilateral hippocampal volume from total gray matter. Partial Spearman associations were estimated after residualizing for ICV, age, sex, and education, then repeated with clinical group included. Duplicate same-day records were resolved using the lower image identifier. Because the FreeSurfer records lacked final visual-QC ratings, these analyses were considered exploratory.

1.13 Diffusion-Protocol and Registration QC Audit

The archived processing environment was MRtrix3 3.0.6 and FSL 6.0.7.18, including FLIRT. Five-tissue-type images were generated with `5ttgen fsl`. Response functions were estimated with `dwi2response dhollander`, and the same `dwi2fod msmt_csd` command, with white-matter, gray-matter, and CSF outputs, was applied to all datasets. This included 266 datasets with one nonzero shell and 15 with three nonzero shells; the use of the same multi-tissue command in the single-nonzero-shell datasets is therefore retained as a processing limitation. The final DWI inputs contained nine total-volume categories (22, 31, 36, 42, 46, 54, 55, 68, and 127), b0-count categories of 1, 4, 5, 6, 7, 8, and 13, and nonzero diffusion-weighted volume-count categories of 16, 30, 32, 36, 41, 48, 60, and 114. Observed shell sets were $b = 0/1000$, $b = 0/1300$, and $b = 0/500/1000/2000$ s/mm^2. Fifteen 127-volume inputs were PA acquisitions, each containing 13 b0 and 114 nonzero diffusion-weighted volumes. Protocol-specific counts are given in Supplementary Table S21.

For retrospective registration QC, the mean b0 image was registered to the first T1-derived 5TT volume using six-degree-of-freedom FLIRT with mutual-information cost and trilinear interpolation. The transform was imported with `transformconvert`; its inverse was applied to the T1-derived brain mask in the mean-b0 grid with linear interpolation. The combined Schaefer–Tian atlas was transferred from MNI to native T1 space using six-degree-of-freedom FLIRT with mutual-information cost and nearest-neighbour label interpolation. Direct overlay inspection confirmed that the resulting native-T1 node image and tractogram shared the same physical coordinate frame. The node image was therefore supplied directly to `tck2connectome`, without an additional voxel-grid resampling step before streamline-to-node assignment. For retrospective QC only, the native-T1 atlas was transformed into the mean-b0 grid with nearest-neighbour interpolation to calculate coverage and generate overlays.

Let M_{DWI} and M_{T1} denote the DWI brain mask and transformed T1 brain mask, and let H denote the transformed bilateral hippocampal-label mask. Brain-mask overlap was quantified as $2|M_{\mathrm{DWI}} \cap M_{\mathrm{T1}}|/(|M_{\mathrm{DWI}}| + |M_{\mathrm{T1}}|)$, and hippocampal coverage as $|H \cap M_{\mathrm{DWI}}|/|H|$. Dice below 0.70 or hippocampal coverage below 0.90 triggered targeted review. Registration montages displayed transformed T1 brain-mask contours over the mean b0 image, and hippocampal montages displayed transformed hippocampal-label contours over the same image. These measures identify gross mask-overlap or out-of-brain placement problems but do not establish anatomical localization accuracy. The audit was completed quantitatively for all 281 participants. The five coverage-trigger cases comprised one CN, two MCI, and two AD participants. Manual inspection confirmed acceptable T1–DWI registration and hippocampal-label placement in all five cases, without gross misregistration or erroneous hippocampal localization. Their exclusion was evaluated in a two-endpoint sensitivity family. Within the final 281-participant analytic cohort, there were no additional registration, tractography/connectome-availability, or QC exclusions. Candidate-level failure counts preceding formation of the final analytic cohort were not available in the archived processing records.

1.14 Motion and Multicenter Sensitivity Analyses

Mean and maximum absolute and relative RMS displacement were derived from each participant's FSL `eddy_movement_rms` output. Group differences were tested using Kruskal–Wallis tests, and the primary controllability models were repeated with mean absolute and relative RMS displacement as additional covariates. Multicenter sensitivity models sequentially adjusted for scanner manufacturer and ADNI phase; site; and, in the strict model, scanner model, phase, site, acquisition acceleration, NIfTI-derived diffusion-volume-count category, slice thickness, and both motion measures. After reconciliation against the final NIfTI inputs, the strict model contained 12 scanner-model, three phase, 43 site, two acceleration, nine volume-count, and four slice-thickness levels; its nominal 76-column

full design had rank 71 and 210 residual degrees of freedom. All controllability models used 10,000 Freedman–Lane permutations. ComBat was not applied because diagnosis, scanner, phase, and site were not fully crossed.

Acquisition overlap was evaluated in three complementary analyses. First, generalized overlap weights were estimated from scanner model, phase, site, acceleration, NIfTI-derived volume-count category, slice thickness, and mean absolute and relative RMS displacement. Balance was assessed using the maximum absolute pairwise standardized mean difference and group-specific effective sample sizes. Second, a connected exact-cell common-support analysis retained exact acquisition cells containing at least two clinical groups, with the additional requirement that every pair of clinical groups shared at least one cell. The model adjusted for age, sex, education, both motion measures, and acquisition-cell indicators; Freedman–Lane residual permutations were restricted within cell. Third, we quantified the feasibility of the still stricter requirement that every retained cell contain all four clinical groups. Because overlap weighting retained material imbalance and no exact cell contained all four groups after protocol reconciliation, these diagnostics and the connected common-support result were interpreted jointly rather than as proof that acquisition confounding had been removed.

1.15 SMC Sample-Size Sensitivity Analyses

Balanced subsampling was repeated 1,000 times with 27 participants per group. The partial F, ΔR^2, significance indicator, and adjusted SMC direction were recorded in each repetition. Each SMC participant was also removed in turn, and the primary 10,000-permutation model was rerun. Because resamples overlapped, these analyses were interpreted as stability checks rather than independent replication.

2 Supplementary Results

2.1 Cortical and Connectivity Visualizations

The descriptive cortical maps were broadly similar across groups, and none of the seven aggregated cortical networks survived BH-FDR correction. Visual variation in Supplementary Fig. S1 therefore should not be interpreted as parcel-level statistical evidence. Supplementary Fig. S2 shows widespread hippocampal and Tian atlas-defined non-cortical structural connectivity with cortical networks; these thresholded displays were descriptive only.

2.2 Rank and Unranked Hippocampal Results

Both matrix-scale-adjusted unranked omnibus effects were retained, but the average-controllability pairwise pattern was scale dependent. The SMC–CN and SMC–MCI differences survived correction only for within-subject ranks, whereas the AD–MCI difference survived correction only in the matrix-scale-adjusted unranked analysis (Supplementary Tables S14 and S15).

2.3 MCI Subtype Results

Within 180 days of the selected diffusion-MRI session, 80 of the 88 MCI participants had sufficient records for subtype classification. Of these, 76 had memory-only features, two had mixed memory and non-memory features, and two had non-memory-only features. Seventy-seven met the stricter memory-feature-plus-objective-impairment definition. Clinical-group effects remained significant after restriction to either amnestic definition (all $p_{\mathrm{FDR}} \leq 0.0012$; Supplementary Table S1).

2.4 Hippocampal Long-Axis Results

Clinical-group effects survived correction in the bilateral medial head, lateral head, body, and tail for both controllability measures, indicating macroscopic long-axis heterogeneity without resolving histological subfields (Supplementary Table S2).

2.5 Education Moderation Results

Diagnosis-by-education interactions were not significant for average or modal controllability (both $p_{\mathrm{FDR}} = 0.748$; Supplementary Table S3).

2.6 Cognitive Association Results

Demographic-adjusted controllability–cognition associations were significant after BH-FDR correction (absolute $\rho = 0.134$–0.256). In the matched ICV-adjusted comparison, hippocampal volume showed larger associations with cognition (absolute $\rho = 0.225$–0.453) than controllability (absolute $\rho = 0.124$–0.253; Supplementary Table S4).

2.7 Vascular and WMH Results

Modified Hachinski scores did not differ among groups and were not associated with the analyzed imaging or cognitive outcomes after correction. Normalized WMH burden differed among groups and, after diagnosis and demographic adjustment, was associated with lower average controllability ($\rho = -0.142$, $p_{\mathrm{FDR}} = 0.044$), higher modal controllability ($\rho = 0.137$, $p_{\mathrm{FDR}} = 0.044$), and lower MoCA ($\rho = -0.247$, $p_{\mathrm{FDR}} = 0.006$; Supplementary Table S5).

2.8 Shared and Incremental Information Results

In the 270-participant exact-date FreeSurfer sample, the harmonized structural models explained 54.6% and 55.0% of raw average- and modal-controllability variance. Adding rank-based average and modal controllability produced an FDR-corrected increment only for CDR memory among the five cognitive outcomes ($\Delta R^2 = 0.0249$, 95% CI 0.0047–0.0638, $p_{\mathrm{FDR}} = 0.0374$). The MMSE increment was nominal before correction but did not survive BH-FDR correction ($\Delta R^2 = 0.0149$, 95% CI 0.0000–0.0530, $p = 0.0418$, $p_{\mathrm{FDR}} = 0.0974$). The multinomial model retained $\Delta R^2_{\mathrm{McFadden}} = 0.0219$ (95% CI 0.0103–0.0606, $p = 0.0189$; Supplementary Table S6).

2.9 Motion Sensitivity Results

None of the four RMS motion summaries differed significantly among groups (all $p_{\mathrm{FDR}} \geq 0.233$), and group effects remained significant after motion adjustment (Supplementary Table S7).

2.10 Multicenter Sensitivity Results

Clinical-group effects remained significant in the strict scanner/site/protocol/motion model after reconciliation to the final NIfTI volume counts (average, $p_{\mathrm{FDR}} = 0.0093$; modal, $p_{\mathrm{FDR}} = 0.0093$; Supplementary Table S8). The full design had rank 71 and 210 residual degrees of freedom. Generalized overlap weighting left a maximum absolute pairwise standardized mean difference of 1.042 and an overall effective sample size of 45.2 (group-specific values, 34.8 CN, 11.5 SMC, 6.7 MCI, and 20.9 AD), indicating inadequate residual balance; weighted estimates were therefore retained only as diagnostics. The connected exact-cell common-support analysis retained 218 participants in 46 cells (CN 93, SMC 24, MCI 66, and AD 35). Group effects remained significant for average rank ($F_{3,164} = 5.663$, $p_{\mathrm{FDR}} = 0.0025$, partial $\eta^2 = 0.094$) and modal rank ($F_{3,164} = 5.112$, $p_{\mathrm{FDR}} = 0.0025$, partial $\eta^2 = 0.086$). No exact acquisition cell contained all four groups. Direct SMC–MCI overlap comprised four cells containing five SMC and six MCI participants; the analysis therefore does not establish that the SMC–MCI classifier is free of acquisition confounding (Supplementary Table S20). The temporally matched diagnostic-record sensitivity analysis did not alter the interpretation of either primary endpoint.

2.11 Registration and Label-Coverage QC Results

The retrospective audit was completed for all 281 participants. Five cases crossed the hippocampal-coverage review threshold (one CN, no SMC, two MCI, and two AD). Manual inspection confirmed acceptable registration and hippocampal-label placement in all five cases. Excluding these cases retained

the average-rank group effect ($F_{3,269} = 5.554$, $p_{\mathrm{FDR}} = 0.0017$, partial $\eta^2 = 0.058$) and modal-rank group effect ($F_{3,269} = 5.264$, $p_{\mathrm{FDR}} = 0.0017$, partial $\eta^2 = 0.055$; Supplementary Table S21).

2.12 SMC Stability Results

Balanced subsampling retained the group effect in 99.9% of average- and 99.7% of modal-controllability repetitions, and both effects remained significant in all leave-one-SMC-out analyses (Supplementary Table S9).

2.13 Global Gray-Matter Results

In 270 FreeSurfer-matched participants, hippocampal volume covaried with total and non-hippocampal gray matter after ICV and demographic adjustment and after additional diagnosis adjustment (Supplementary Table S10).

2.14 Amyloid PET Results

Within 365 days, binary amyloid status was available for 230 participants and Centiloids for 226. Diagnosis effects remained significant after adjustment for either measure, whereas neither amyloid measure was independently associated with controllability after diagnosis adjustment. The 730-day analysis gave the same conclusion (Supplementary Table S11).

2.15 Classification Results

Complete low-dimensional results are reported in Table II . Eleven of 24 complete-pipeline permutation tests remained significant after Holm correction. Bilateral volume had the highest AUC point estimates for CN versus AD (0.820) and MCI versus AD (0.825). For SMC versus MCI, the combined feature set had the highest estimate (0.729), while left controllability had the highest estimate among individual feature sets (0.701). In the common 112-person sample, none of the SMC-versus-MCI candidate-versus-volume differences survived Holm correction (Supplementary Table S16). The classification results therefore describe task-specific point-estimate ordering without evidence that controllability is statistically superior to volume. The high-dimensional connectivity AUCs are retained as descriptive estimates only (Supplementary Table S12).

3 Supplementary Tables

Table S1: MCI subtype audit and aMCI-restricted sensitivity analyses.

Matching window or model	Usable subtype record	Memory only	Mixed	Non-memory only	Strict aMCI	Unclassified
A. MCI subtype audit						
Within 180 days	80	76	2	2	77	8
Within 365 days	84	80	2	2	81	4
Any recorded MCI visit	88	83	2	3	84	0

MCI inclusion model	Measure	n_{MCI}	Total n	F	p_{perm}	p_{FDR}
B. Clinical-group sensitivity analyses						
All clinically defined MCI	Average	88	281	6.495	0.0004	0.0007
	Modal	88	281	6.041	0.0007	0.0007
Memory-feature aMCI within 180 days	Average	78	271	6.485	0.0011	0.0012
	Modal	78	271	6.061	0.0012	0.0012
Memory feature + objective impairment	Average	77	270	6.584	0.0004	0.0006
	Modal	77	270	6.181	0.0006	0.0006

Note: The first numeric column reports availability of the records required for subtype classification within 180 days of the selected diffusion-MRI session. Memory only denotes an amnestic-only `DXMDES` record; mixed denotes both memory and non-memory features. Strict aMCI required a memory-feature `DXMDES` record and `DXMPTR5`=Yes, indicating objective memory impairment for age and education. Panel B reports Freedman–Lane permutation F-tests with 10,000 permutations adjusted for age, sex, and education. BH-FDR correction was applied across average and modal controllability within each inclusion model. These subtype analyses do not establish subsequent conversion to AD dementia.

Table S2: Clinical-group effects across Tian S3 hippocampal long-axis subdivisions.

Metric	Bilateral subdivision	n	$F(3, 274)$	p_{perm}	p_{FDR}
Average	Medial head	281	6.712	0.0007	0.0024
Average	Lateral head	281	5.693	0.0012	0.0024
Average	Body	281	3.825	0.0111	0.0148
Average	Tail	281	3.387	0.0183	0.0183
Modal	Medial head	281	6.566	0.0002	0.0008
Modal	Lateral head	281	5.009	0.0025	0.0050
Modal	Body	281	3.816	0.0099	0.0132
Modal	Tail	281	2.735	0.0455	0.0455

Note: Omnibus clinical-group effects were tested after adjustment for age, sex, and education using 10,000 Freedman–Lane permutations. BH-FDR correction was applied across the four subdivisions separately within average and modal controllability. Tian S3 long-axis parcels are macroscopic connectivity-based subdivisions and are not equivalent to histological hippocampal subfields.

Table S3: Exploratory diagnosis-by-education interaction analyses for hippocampal controllability.

Outcome	n	$F(3, 271)$	p_{perm}	p_{FDR}	ΔR^2
Average controllability	281	0.415	0.737	0.748	0.004
Modal controllability	281	0.403	0.748	0.748	0.004

Note: Models included age, sex, education, diagnosis, and the three diagnosis-by-education interaction terms, with CN as the reference group. The reported F-test compares models with and without the interaction block using 10,000 Freedman–Lane permutations. Education and age were standardized. BH-FDR correction was applied across the two controllability outcomes.

Table S4: Partial Spearman correlations of hippocampal controllability and volume with cognitive performance.

A. Primary controllability–cognition associations

Cognitive measure	n	Average controllability ρ (p_{FDR})	Modal controllability ρ (p_{FDR})
MMSE	279	0.146 (0.021)	−0.134 (0.025)
ADAS-Cog 11	264	−0.189 (0.005)	0.180 (0.006)
ADAS-Cog 13	264	−0.194 (0.005)	0.180 (0.006)
CDR memory	280	−0.256 (< 0.001)	0.243 (< 0.001)
MoCA	169	0.183 (0.021)	−0.176 (0.025)

B. Matched ICV-adjusted comparison with hippocampal volume

Cognitive measure	n	Average controllability ρ (p_{FDR})	Modal controllability ρ (p_{FDR})	Hippocampal volume ρ (p_{FDR})
MMSE	268	0.139 (0.024)	−0.124 (0.043)	0.367 (< 0.001)
ADAS-Cog 11	254	−0.196 (0.003)	0.183 (0.005)	−0.437 (< 0.001)
ADAS-Cog 13	254	−0.204 (0.002)	0.187 (0.005)	−0.453 (< 0.001)
CDR memory	269	−0.253 (< 0.001)	0.236 (< 0.001)	−0.393 (< 0.001)
MoCA	166	0.195 (0.015)	−0.188 (0.018)	0.225 (0.005)

Note: Panel A residualized the cognitive and controllability variables for age, sex, and education. BH-FDR correction was applied across the 10 controllability–cognition associations in Panel A. Panel B used the same participants for all three imaging measures and additionally adjusted for ICV; BH-FDR correction was applied across its 15 comparisons. Higher MMSE and MoCA scores indicate better cognition, whereas higher ADAS-Cog and CDR memory scores indicate greater impairment. Hippocampal volume is the sum of the left and right FreeSurfer 7.x volumes.

Table S5: Associations of Modified Hachinski score and white-matter hyperintensity burden with hippocampal imaging features and cognition.

Analysis or predictor	Outcome or coverage	n	Statistic	p	p_{FDR}
A. Data coverage					
CN	Hachinski / WMH / both	110	88 / 110 / 88	–	–
SMC	Hachinski / WMH / both	27	21 / 27 / 21	–	–
MCI	Hachinski / WMH / both	88	67 / 87 / 67	–	–
AD	Hachinski / WMH / both	56	52 / 54 / 51	–	–
Overall	Hachinski / WMH / both	281	228 / 278 / 227	–	–
B. Clinical-group differences					
Modified Hachinski score	Four-group comparison	228	$H = 4.329$	0.228	0.228
Normalized WMH burden	Four-group comparison	278	$H = 27.490$	< 0.0001	< 0.0001
C. Diagnosis- and demographic-adjusted partial Spearman associations					
Modified Hachinski score	Average controllability	228	$\rho = 0.071$	0.286	0.692
Modified Hachinski score	Modal controllability	228	$\rho = -0.063$	0.346	0.692
Modified Hachinski score	Hippocampal strength	228	$\rho = 0.039$	0.560	0.746
Modified Hachinski score	Hippocampal betweenness	228	$\rho = 0.009$	0.893	0.893
Modified Hachinski score	MMSE	227	$\rho = -0.084$	0.206	0.846
Modified Hachinski score	ADAS-Cog 11	212	$\rho = 0.013$	0.846	0.846
Modified Hachinski score	ADAS-Cog 13	212	$\rho = 0.029$	0.674	0.846
Modified Hachinski score	CDR memory	227	$\rho = 0.043$	0.516	0.846
Modified Hachinski score	MoCA	130	$\rho = -0.034$	0.704	0.846
Modified Hachinski score	Normalized WMH burden	227	$\rho = 0.004$	0.949	0.949
Normalized WMH burden	Average controllability	278	$\rho = -0.142$	0.018	0.044
Normalized WMH burden	Modal controllability	278	$\rho = 0.137$	0.022	0.044
Normalized WMH burden	Hippocampal strength	278	$\rho = -0.110$	0.067	0.089
Normalized WMH burden	Hippocampal betweenness	278	$\rho = -0.062$	0.304	0.304
Normalized WMH burden	MMSE	276	$\rho = -0.122$	0.043	0.108
Normalized WMH burden	ADAS-Cog 11	262	$\rho = 0.082$	0.186	0.229
Normalized WMH burden	ADAS-Cog 13	262	$\rho = 0.075$	0.229	0.229
Normalized WMH burden	CDR memory	277	$\rho = 0.104$	0.084	0.140
Normalized WMH burden	MoCA	167	$\rho = -0.247$	0.0013	0.006

Note: Modified Hachinski scores and UCD white-matter hyperintensity (WMH) measurements were matched to the selected diffusion MRI session within 365 days. WMH burden was calculated as $100 \times$ TOTAL_WMH/CEREBRUM_TCV and transformed using $\log(1 + x)$. Partial Spearman associations controlled for clinical group, age, sex, and education. BH-FDR correction was applied separately within each vascular predictor and outcome family.

Table S6: Harmonized shared and incremental information analyses using exact-date FreeSurfer 7.2.0 hippocampal volume and intracranial volume.

A. Variance in unranked controllability explained by volume and topology

Outcome	n	Model R^2 (95% CI)	Adjusted R^2
Raw average controllability	270	0.546 (0.449–0.688)	0.533
Raw modal controllability	270	0.550 (0.481–0.655)	0.536

B. Increment after jointly adding average- and modal-controllability ranks

Outcome	n	Base R^2	Full R^2	ΔR^2 (95% CI)	Partial F	df	p	p_{FDR}
MMSE	268	0.388	0.403	0.0149 (0.0000–0.0530)	3.213	2,257	0.0418	0.0974
ADAS-Cog 11	254	0.421	0.431	0.0097 (0.0000–0.0364)	2.079	2,243	0.1273	0.1273
ADAS-Cog 13	254	0.444	0.454	0.0101 (0.0000–0.0359)	2.257	2,243	0.1068	0.1273
CDR memory	269	0.332	0.357	0.0249 (0.0047–0.0638)	4.991	2,258	0.0075	0.0374
MoCA	166	0.322	0.347	0.0244 (0.0000–0.0758)	2.893	2,155	0.0584	0.0974

C. Increment for multinomial clinical-group membership

Outcome	n	Base McFadden R^2	Full McFadden R^2	$\Delta R^2_{\mathrm{McFadden}}$ (95% CI)	χ^2	df	p
Clinical-group membership	270	0.1885	0.2105	0.0219 (0.0103–0.0606)	15.180	6	0.0189

D. Availability of exact-date FreeSurfer variables

Group	Cohort	FS complete	FS missing	MMSE	ADAS-11	ADAS-13	CDR memory	MoCA
CN	110	105	5	104	104	104	105	77
SMC	27	27	0	27	27	27	27	3
MCI	88	85	3	84	73	73	84	67
AD	56	53	3	53	50	50	53	19
Total	281	270	11	268	254	254	269	166

Note: ICV, left hippocampal volume, and right hippocampal volume were `ST10CV`, `ST29SV`, and `ST88SV` from `UCSFFSX7_11Jul2026.csv`. Eligible 3.0 T records exactly matched the diffusion-MRI date; lower numeric `IMAGEUID` resolved duplicates. All included records used FreeSurfer 7.2.0. `STATUS` was `partial`; `OVERALLQC` and `HIPPOQC` were not populated. Panel A base models included age, sex, education, ICV, left and right hippocampal volumes as separate predictors, strength, and betweenness. Panel B jointly added rank-based average and modal controllability and used 5,000 group-stratified percentile bootstrap resamples; the intervals are descriptive, whereas inference used the BH-FDR-corrected partial F-tests across five outcomes. Panel C used a multinomial likelihood-ratio test and a 5,000-resample group-stratified percentile-bootstrap interval.

Table S7: Eddy-derived motion and motion-adjusted sensitivity analyses of hippocampal controllability.

Adjustment model or test	Measure	n	Statistic	p value	p_{FDR}
A. Clinical-group differences in eddy-derived motion					
Kruskal–Wallis group test	Mean absolute RMS	281	$H = 4.347$	0.226	0.285
Kruskal–Wallis group test	Mean relative RMS	281	$H = 6.448$	0.092	0.233
Kruskal–Wallis group test	Maximum absolute RMS	281	$H = 3.787$	0.285	0.285
Kruskal–Wallis group test	Maximum relative RMS	281	$H = 5.903$	0.116	0.233
B. Sensitivity analyses after motion adjustment					
Demographic covariates	Average controllability	281	$F = 6.495$	0.0004	0.0007
	Modal controllability	281	$F = 6.041$	0.0007	0.0007
Demographic covariates + motion	Average controllability	281	$F = 6.237$	0.0006	0.0012
	Modal controllability	281	$F = 5.786$	0.0014	0.0014

Note: RMS displacement measures were obtained from FSL `eddy` outputs. Panel A reports Kruskal–Wallis tests, with BH-FDR correction across the four motion summaries. Panel B reports Freedman–Lane permutation F-tests with 10,000 permutations. All Panel B models included age, sex, and education; the motion-adjusted model additionally included mean absolute and mean relative RMS displacement. BH-FDR correction was applied across average and modal controllability within each Panel B model. RMS, root mean square.

Table S8: Multicenter and acquisition-protocol sensitivity analyses of hippocampal controllability.

Adjustment model	**Measure**	n	F	p_{perm}	p_{FDR}
Scanner manufacturer + ADNI phase	Average controllability	281	5.362	0.0022	0.0026
	Modal controllability	281	4.852	0.0026	0.0026
Scanner manufacturer + phase + site	Average controllability	281	3.792	0.0119	0.0166
	Modal controllability	281	3.522	0.0166	0.0166
Scanner/site/protocol/motion model	Average controllability	281	4.368	0.0057	0.0093
	Modal controllability	281	3.977	0.0093	0.0093

Note: All models included age, sex, and education and were evaluated using Freedman–Lane permutation F-tests with 10,000 permutations. After reconciliation to the final NIfTI inputs, the strict scanner/site/protocol/motion model additionally included 12 scanner-model levels, three ADNI phases, 43 sites, two acceleration levels, nine NIfTI-derived diffusion-volume-count levels, four slice-thickness levels, and mean absolute and relative RMS displacement. Its nominal 76-column full design had rank 71 and 210 residual degrees of freedom. BH-FDR correction was applied across average and modal controllability within each adjustment model. ComBat was not applied because diagnosis, ADNI phase, scanner, and site were not fully crossed.

Table S9: SMC sample-size and unequal-group sensitivity analyses of hippocampal controllability.

Analysis	**Measure**	**Partial F**	**ΔR^2**	**Significant runs**	**SMC pattern retained**
Balanced subsampling	Average	5.663 (3.439–9.859)	0.131 (0.084–0.212)	99.9%	99.9%
	Modal	5.349 (3.295–9.588)	0.127 (0.081–0.205)	99.7%	99.9%
Leave-one-SMC-out	Average	5.960–6.863	0.058–0.066	100%	100%
	Modal	5.518–6.398	0.054–0.062	100%	100%

Note: Balanced subsampling comprised 1,000 repetitions with 27 participants per group; values are medians (2.5th–97.5th percentiles), and "significant runs" denotes the percentage with parametric partial F-test $p < 0.05$. Leave-one-SMC-out comprised 27 analyses with 26 SMC participants and all CN, MCI, and AD participants retained; values are ranges across iterations, all of which remained significant by Freedman–Lane tests with 10,000 permutations (average controllability, $p = 0.0002$–0.0010; modal controllability, $p = 0.0003$–0.0017). The SMC pattern was defined as the highest adjusted mean average controllability and lowest adjusted mean modal controllability among the four groups. Overlapping resamples are stability checks rather than independent replications.

Table S10: Same-day FreeSurfer global gray-matter sensitivity analysis.

A. Same-day coverage

Group	Cohort n	Matched n	Coverage
CN	110	105	95.5%
SMC	27	27	100.0%
MCI	88	85	96.6%
AD	56	53	94.6%
Overall	281	270	96.1%

B. Partial Spearman associations

Hippocampal measure	Comparison measure	n	ρ	p_{FDR}
Bilateral hippocampus	Total gray matter	270	0.498	7.54×10^{-18}
Bilateral hippocampus	Non-hippocampal gray matter	270	0.480	1.06×10^{-16}
Bilateral hippocampus	Cortical gray matter	270	0.468	5.01×10^{-16}
Bilateral hippocampus	Subcortical gray matter	270	0.595	1.91×10^{-26}
Left hippocampus	Non-hippocampal gray matter	270	0.470	4.48×10^{-16}
Right hippocampus	Non-hippocampal gray matter	270	0.440	3.05×10^{-14}

C. Associations additionally adjusted for diagnostic group

Hippocampal measure	Comparison measure	n	ρ	p_{FDR}
Bilateral hippocampus	Total gray matter	270	0.428	5.67×10^{-13}
Bilateral hippocampus	Non-hippocampal gray matter	270	0.412	3.38×10^{-12}
Bilateral hippocampus	Cortical gray matter	270	0.395	2.06×10^{-11}
Bilateral hippocampus	Subcortical gray matter	270	0.537	7.86×10^{-21}
Left hippocampus	Non-hippocampal gray matter	270	0.404	7.63×10^{-12}
Right hippocampus	Non-hippocampal gray matter	270	0.366	5.52×10^{-10}

Note: Panel B associations controlled for ICV, age, sex, and education. Panel C additionally included three diagnostic-group indicator variables. Non-hippocampal gray matter was calculated by subtracting bilateral hippocampal volume from total gray-matter volume, avoiding part–whole correlation. BH-FDR correction was applied across the six associations within each panel. These are cross-sectional volume associations, not longitudinal atrophy rates. The matched FreeSurfer records lacked final visual QC and these results are exploratory.

Table S11: Amyloid PET sensitivity analyses for hippocampal controllability.

Window	Amyloid variable	Tested effect	Outcome	n	F	p (p_{FDR})
365 d	Status	Diagnosis	Average	230	$F_{3,222} = 5.521$	0.0018 (0.0018)
365 d	Status	Diagnosis	Modal	230	$F_{3,222} = 5.321$	0.0016 (0.0018)
365 d	Status	Amyloid	Average	230	$F_{1,222} = 0.327$	0.5689 (0.5689)
365 d	Status	Amyloid	Modal	230	$F_{1,222} = 0.392$	0.5381 (0.5689)
365 d	Centiloids	Diagnosis	Average	226	$F_{3,218} = 5.483$	0.0009 (0.0018)
365 d	Centiloids	Diagnosis	Modal	226	$F_{3,218} = 5.160$	0.0021 (0.0021)
365 d	Centiloids	Amyloid	Average	226	$F_{1,218} = 0.877$	0.3463 (0.3675)
365 d	Centiloids	Amyloid	Modal	226	$F_{1,218} = 0.781$	0.3675 (0.3675)
730 d	Status	Diagnosis	Average	241	$F_{3,233} = 6.228$	0.0004 (0.0008)
730 d	Status	Diagnosis	Modal	241	$F_{3,233} = 6.100$	0.0008 (0.0008)
730 d	Status	Amyloid	Average	241	$F_{1,233} = 0.023$	0.8797 (0.8797)
730 d	Status	Amyloid	Modal	241	$F_{1,233} = 0.051$	0.8246 (0.8797)
730 d	Centiloids	Diagnosis	Average	237	$F_{3,229} = 6.106$	0.0006 (0.0008)
730 d	Centiloids	Diagnosis	Modal	237	$F_{3,229} = 5.850$	0.0008 (0.0008)
730 d	Centiloids	Amyloid	Average	237	$F_{1,229} = 0.394$	0.5188 (0.5628)
730 d	Centiloids	Amyloid	Modal	237	$F_{1,229} = 0.336$	0.5628 (0.5628)

Note: The diagnosis effect was tested after adjustment for the listed amyloid variable, age, sex, and education. The amyloid effect was tested after adjustment for diagnosis, age, sex, and education. Status and Centiloids were analyzed in separate complete-case models, producing different sample sizes; valid negative Centiloid values were retained. All p values were obtained using 10,000 Freedman–Lane permutations. BH-FDR correction was applied across average and modal controllability within each tested effect.

Table S12: Classification performance based on hippocampal structural connectivity features in four binary tasks.

Task	AUC (95% CI)	Interpretation
CN vs AD	0.601 (0.508–0.691)	Descriptive exploratory estimate
SMC vs MCI	0.693 (0.584–0.797)	Descriptive exploratory estimate
MCI vs AD	0.634 (0.540–0.731)	Descriptive exploratory estimate
CN vs SMC	0.680 (0.573–0.780)	Descriptive exploratory estimate

Note: AUC confidence intervals were estimated using 2,000 bootstrap resamples of subject-level out-of-sample predictions from nested cross-validation. The previous fixed-score permutation values were removed because the full high-dimensional pipeline was not refitted under label permutation. CN, cognitively normal; SMC, subjective memory complaints; MCI, mild cognitive impairment; AD, Alzheimer's disease.

Table S13: Primary rank endpoints and matrix-scale-adjusted unranked hippocampal controllability sensitivity analyses.

A. Adjusted group estimates (95% CI)

Scale	Metric	CN	SMC	MCI	AD
Within-subject rank	Average	345.3 (329.5–360.3)	383.9 (366.1–399.6)	324.7 (306.7–341.5)	307.4 (286.7–327.7)
Within-subject rank	Modal	107.9 (92.5–124.0)	66.5 (49.4–85.0)	128.8 (111.5–148.0)	142.4 (122.5–163.9)
Unranked raw value	Average	1.173 (1.152–1.195)	1.218 (1.174–1.265)	1.169 (1.140–1.204)	1.116 (1.096–1.139)
Unranked raw value	Modal	0.905 (0.894–0.914)	0.877 (0.858–0.895)	0.913 (0.901–0.924)	0.929 (0.918–0.940)

B. Omnibus clinical-group effects

Analysis	Metric	F	df	p_{perm}	p_{FDR}	Partial η^2
Primary rank	Average	6.495	3,274	0.0004	0.0028	0.066 (0.034–0.127)
Primary rank	Modal	6.041	3,274	0.0007	0.0049	0.062 (0.032–0.121)
Raw, matrix-scale adjusted	Average	4.426	3,273	0.0066	0.0066	0.046
Raw, matrix-scale adjusted	Modal	6.313	3,273	0.0004	0.0008	0.065
Raw, matrix-scale+density adjusted	Average	4.232	3,272	0.0062	0.0062	0.045
Raw, matrix-scale+density adjusted	Modal	5.306	3,272	0.0016	0.0032	0.055

C. Direct average–modal association

Scale and adjustment	n	Partial Spearman ρ (95% CI)
Within-subject ranks; age, sex, education	281	−0.992 (−0.994 to −0.987)
Unranked; age, sex, education, matrix scale	281	−0.930 (−0.955 to −0.896)

Note: Adjusted estimates and confidence intervals used 5,000 group-stratified bootstrap resamples. Omnibus tests used 10,000 Freedman–Lane permutations. Primary rank values retain correction across seven Tian structures within endpoint; each unranked omnibus family contains average and modal outcomes. Matrix scale is $\log(1 + W_{\text{total}})$, where $W_{\text{total}} = \sum_{i<j} A_{ij}$ for the final inverse-node-volume-scaled matrix before spectral stabilization.

Table S14: Multiplicity-controlled pairwise contrasts for the primary hippocampal within-subject rank endpoints.

Endpoint	Contrast	Adjusted difference (95% CI)	Standardized difference	p	p_{FDR}
Average rank	SMC–CN	38.6 (15.5, 60.9)	0.49	0.0225	0.0337
	MCI–CN	−20.6 (−43.7, 2.0)	−0.26	0.0733	0.0977
	AD–CN	−37.9 (−63.9, −12.4)	−0.48	0.0049	0.0118
	MCI–SMC	−59.2 (−83.3, −34.3)	−0.75	0.0011	0.0033
	AD–SMC	−76.5 (−102.6, −50.0)	−0.97	0.0002	0.0018
	AD–MCI	−17.3 (−43.6, 8.8)	−0.22	0.2027	0.2211
Modal rank	SMC–CN	−41.5 (−65.1, −17.2)	−0.51	0.0196	0.0336
	MCI–CN	20.9 (−2.3, 45.6)	0.26	0.0823	0.0988
	AD–CN	34.5 (8.8, 61.3)	0.43	0.0126	0.0252
	MCI–SMC	62.3 (36.4, 88.2)	0.77	0.0007	0.0028
	AD–SMC	76.0 (48.5, 102.3)	0.94	0.0003	0.0018
	AD–MCI	13.6 (−13.5, 40.5)	0.17	0.3317	0.3317

Note: Differences are first-named minus second-named group. Confidence intervals used 5,000 group-stratified bootstrap resamples and are individually unadjusted. The 12 two-sided permutation tests formed one BH-FDR family.

Table S15: Multiplicity-controlled pairwise contrasts for matrix-scale-adjusted unranked hippocampal controllability.

Metric	Contrast	Adjusted difference (95% CI)	Standardized difference	p_{FDR}
Average	SMC–CN	0.0446 (−0.0059, 0.0966)	0.36	0.1193
	MCI–CN	−0.0040 (−0.0408, 0.0350)	−0.03	0.8259
	AD–CN	−0.0569 (−0.0879, −0.0244)	−0.46	0.0187
	MCI–SMC	−0.0486 (−0.1032, 0.0078)	−0.39	0.1108
	AD–SMC	−0.1016 (−0.1532, −0.0529)	−0.82	0.0048
	AD–MCI	−0.0529 (−0.0947, −0.0148)	−0.43	0.0255
Modal	SMC–CN	−0.0272 (−0.0485, −0.0058)	−0.53	0.0255
	MCI–CN	0.0083 (−0.0073, 0.0236)	0.16	0.2775
	AD–CN	0.0245 (0.0092, 0.0404)	0.48	0.0135
	MCI–SMC	0.0356 (0.0133, 0.0579)	0.70	0.0092
	AD–SMC	0.0517 (0.0294, 0.0742)	1.01	0.0012
	AD–MCI	0.0161 (0.0003, 0.0327)	0.32	0.1012

Note: Models additionally adjusted for $\log(1 + W_{\mathrm{total}})$, where $W_{\mathrm{total}} = \sum_{i<j} A_{ij}$ for the final inverse-node-volume-scaled matrix before spectral stabilization. The 12 two-sided permutation tests formed one BH-FDR family; individual bootstrap confidence intervals are unadjusted.

Table S16: Direct paired AUC comparisons of candidate feature sets with bilateral hippocampal volume in common complete-case samples.

Task	Candidate	n	Candidate AUC	Volume AUC	ΔAUC (95% CI)	p_{Holm}
CN vs AD	Global controllability	158	0.653	0.820	−0.167 (−0.281, −0.057)	0.042
	Left controllability	158	0.673	0.820	−0.147 (−0.256, −0.039)	0.120
	Right controllability	158	0.587	0.820	−0.233 (−0.344, −0.120)	0.007
	Topology	158	0.549	0.820	−0.271 (−0.384, −0.148)	0.004
	Combined	158	0.813	0.820	−0.007 (−0.037, 0.025)	1.000
SMC vs MCI	Global controllability	112	0.708	0.654	0.054 (−0.091, 0.199)	1.000
	Left controllability	112	0.675	0.654	0.022 (−0.122, 0.167)	1.000
	Right controllability	112	0.708	0.654	0.055 (−0.098, 0.200)	1.000
	Topology	112	0.617	0.654	−0.037 (−0.196, 0.119)	1.000
	Combined	112	0.729	0.654	0.076 (−0.014, 0.169)	1.000
MCI vs AD	Global controllability	138	0.630	0.825	−0.195 (−0.319, −0.067)	0.019
	Left controllability	138	0.649	0.825	−0.176 (−0.292, −0.058)	0.033
	Right controllability	138	0.454	0.825	−0.370 (−0.499, −0.238)	0.004
	Topology	138	0.490	0.825	−0.335 (−0.456, −0.212)	0.004
	Combined	138	0.808	0.825	−0.016 (−0.042, 0.006)	1.000
CN vs SMC	Global controllability	132	0.606	0.635	−0.029 (−0.194, 0.138)	1.000
	Left controllability	132	0.577	0.635	−0.057 (−0.214, 0.105)	1.000
	Right controllability	132	0.632	0.635	−0.003 (−0.165, 0.163)	1.000
	Topology	132	0.672	0.635	0.037 (−0.119, 0.199)	1.000
	Combined	132	0.697	0.635	0.062 (−0.028, 0.153)	1.000

Note: Candidate and volume models used identical subjects, cross-validation splits, and the same random seed as Table II. Confidence intervals used 5,000 group-stratified resamples of fixed averaged OOF scores and therefore do not include model-refitting variability. Paired score-swap tests were Holm corrected across 20 comparisons.

Table S17: Reconstructed query inventory and partial cohort-selection audit from the ADNI exports and analysis archive.

Stage	Individuals	Sessions/series	Rule or audit status
MRIQC export: all imaging-series rows	3,194	92,098	Imaging-series export used for analysis
3.0 T diffusion-MRI query	1,860	5,793	dMRI and 3.0 T
Unique participant-date sessions	1,860	4,566	Same-date series collapsed; lower image ID broke ties
Eligibility and deterministic session selection	–	–	Complete gradients/phase encoding, matched T1, usable QC, successful registration/tractography/connectome; earliest eligible date
Previous revised cohort	240	240	CN 110, SMC 18, MCI 88, AD 24; all retained
Additional eligible participants	41	41	SMC 9 and AD 32; same eligibility criteria
Final primary cohort	281	281	One selected session per participant; CN 110, SMC 27, MCI 88, AD 56

Note: A session was a unique participant-date combination. Same-date diffusion series were resolved using the lower numeric image identifier; when several eligible dates remained, the earliest eligible date was selected. The final cohort retained the previous 240 participants and added 41 eligible participants identified during the re-audit. The query and final session inventory were recoverable, whereas intermediate historical exclusion counts and reasons were not fully preserved; the table is therefore labeled as a partial audit rather than a complete prospective flow.

Table S18: Cross-classification of clinical group by ADNI phase, site, scanner, and principal diffusion-protocol characteristics.

Phase	Site	Manufacturer	Model	Acceleration	NIfTI volumes	Slice (mm)	CN	SMC	MCI	AD	Total
ADNI3	168	Siemens	Prisma_fit	Accelerated	55	2.0	5	0	7	7	19
ADNI3	135	GE	DISCOVERY MR750w	Accelerated	36	2.0	4	0	8	4	16
ADNIGO/ADNI2	127	GE	DISCOVERY MR750	Accelerated	46	2.7	2	3	0	6	11
ADNIGO/ADNI2	27	GE	DISCOVERY MR750	Accelerated	46	2.7	2	6	0	1	9
ADNI3	3	Siemens	Prisma	Accelerated	55	2.0	4	0	5	0	9
ADNI3	35	Siemens	Prisma_fit	Accelerated	55	2.0	4	0	2	1	7
ADNI3	127	GE	DISCOVERY MR750	Accelerated	54	2.0	4	0	1	2	7
ADNI3	141	Siemens	Verio	Unaccelerated	31	2.0	5	0	2	0	7
ADNI3	109	Siemens	TrioTim	Accelerated	31	2.0	0	0	7	0	7
ADNIGO/ADNI2	16	GE	Signa HDxt	Accelerated	46	2.7	0	0	0	7	7
ADNI3	116	Siemens	TrioTim	Accelerated	31	2.0	4	0	1	1	6
ADNIGO/ADNI2	3	GE	Signa HDxt	Accelerated	55	2.7	3	1	2	0	6
ADNI3	14	Siemens	Verio	Accelerated	31	2.0	4	0	2	0	6
ADNI3	37	Siemens	Prisma	Accelerated	55	2.0	3	0	3	0	6
ADNI3	41	Siemens	Prisma_fit	Accelerated	55	2.0	3	0	3	0	6
ADNI3	129	GE	DISCOVERY MR750	Accelerated	54	2.0	2	0	2	1	5
ADNIGO/ADNI2	3	GE	Signa HDxt	Accelerated	46	2.7	2	1	0	2	5
ADNI3	82	Siemens	Verio	Accelerated	31	2.0	4	0	1	0	5
ADNI3	13	Siemens	Biograph_mMR	Accelerated	31	2.0	1	0	2	1	4
ADNIGO/ADNI2	29	GE	Signa HDxt	Accelerated	46	2.7	1	2	0	1	4
ADNI3	11	Siemens	Prisma_fit	Accelerated	55	2.0	3	0	1	0	4
ADNI3	12	Siemens	Prisma	Accelerated	55	2.0	0	2	2	0	4
ADNI3	27	GE	DISCOVERY MR750	Accelerated	54	2.0	1	0	3	0	4
ADNI3	68	Siemens	Prisma_fit	Accelerated	55	2.0	3	0	1	0	4
ADNI3	70	Siemens	TrioTim	Accelerated	31	2.0	2	0	2	0	4
ADNIGO/ADNI2	7	GE	Signa HDxt	Accelerated	127	2.7	3	0	1	0	4
ADNIGO/ADNI2	98	GE	DISCOVERY MR750	Accelerated	46	2.7	2	0	0	2	4
ADNI3	137	Siemens	Skyra	Accelerated	55	2.0	0	0	4	0	4
ADNI3	9	GE	DISCOVERY MR750w	Accelerated	36	2.0	2	0	1	0	3
ADNI3	16	Siemens	Skyra	Accelerated	55	2.0	2	0	1	0	3
ADNI3	20	Siemens	Prisma_fit	Accelerated	55	2.0	2	1	0	0	3
ADNI3	22	Siemens	TrioTim	Accelerated	31	2.0	0	0	2	1	3
ADNI3	33	Siemens	Skyra	Accelerated	55	2.0	2	0	1	0	3
ADNI3	98	GE	SIGNA Premier	Accelerated	54	2.0	1	0	2	0	3
ADNIGO/ADNI2	21	GE	DISCOVERY MR750	Accelerated	68	2.7	1	2	0	0	3
ADNIGO/ADNI2	27	GE	DISCOVERY MR750	Accelerated	54	2.7	2	1	0	0	3
ADNIGO/ADNI2	126	GE	DISCOVERY MR750	Accelerated	46	2.7	1	0	0	2	3
ADNIGO/ADNI2	127	GE	DISCOVERY MR750	Accelerated	68	2.7	1	2	0	0	3
ADNI3	32	Siemens	Prisma_fit	Accelerated	55	2.0	3	0	0	0	3
ADNIGO/ADNI2	5	GE	DISCOVERY MR750	Accelerated	46	2.7	0	0	0	3	3
ADNIGO/ADNI2	7	GE	Signa HDxt	Accelerated	46	2.7	0	0	0	3	3
ADNI3	2	Siemens	Prisma_fit	Accelerated	55	2.0	1	0	1	0	2
ADNI3	24	Siemens	Prisma_fit	Accelerated	127	2.0	1	0	1	0	2
ADNI3	37	Siemens	Prisma	Accelerated	127	2.0	1	0	1	0	2
ADNI3	114	Siemens	Verio	Unaccelerated	31	2.0	1	0	1	0	2
ADNI3	126	GE	DISCOVERY MR750	Accelerated	54	2.0	1	0	0	1	2
ADNIGO/ADNI2	21	GE	DISCOVERY MR750	Accelerated	54	2.7	1	1	0	0	2
ADNIGO/ADNI2	27	GE	Signa HDxt	Accelerated	46	2.7	0	0	1	1	2
ADNIGO/ADNI2	94	GE	Signa HDxt	Accelerated	46	2.7	1	0	0	1	2

Continued on next page

Table S18 – continued

Phase	Site	Manufacturer	Model	Acceleration	NIfTI volumes	Slice (mm)	CN	SMC	MCI	AD	Total
ADNIGO/ADNI2	99	GE	DISCOVERY MR750	Accelerated	54	2.0	1	0	1	0	2
ADNIGO/ADNI2	126	GE	DISCOVERY MR750	Accelerated	68	2.7	0	1	1	0	2
ADNIGO/ADNI2	127	GE	DISCOVERY MR750	Accelerated	54	2.7	0	1	1	0	2
ADNI3	21	GE	DISCOVERY MR750	Accelerated	54	2.0	2	0	0	0	2
ADNIGO/ADNI2	27	GE	DISCOVERY MR750w	Accelerated	46	2.7	0	0	0	2	2
ADNIGO/ADNI2	57	GE	DISCOVERY MR750	Accelerated	46	2.7	0	2	0	0	2
ADNIGO/ADNI2	129	GE	SIGNA HDx	Accelerated	54	2.7	2	0	0	0	2
ADNI3	2	Siemens	Prisma_fit	Accelerated	127	2.0	0	0	1	0	1
ADNI3	5	GE	DISCOVERY MR750	Accelerated	54	2.0	0	0	1	0	1
ADNI3	21	GE	SIGNA UHP	Accelerated	36	2.0	1	0	0	0	1
ADNI3	29	GE	DISCOVERY MR750w	Accelerated	36	2.0	0	0	1	0	1
ADNI3	36	Siemens	Skyra	Accelerated	55	2.0	0	0	1	0	1
ADNI3	36	Siemens	Skyra	Accelerated	127	2.0	1	0	0	0	1
ADNI3	94	Siemens	Prisma	Accelerated	55	2.0	1	0	0	0	1
ADNI3	127	GE	DISCOVERY MR750	Accelerated	127	2.0	0	0	1	0	1
ADNI3	128	GE	DISCOVERY MR750w	Accelerated	36	2.0	1	0	0	0	1
ADNI3	129	GE	DISCOVERY MR750	Accelerated	22	2.0	0	0	0	1	1
ADNI3	135	GE	SIGNA Premier	Accelerated	36	2.0	0	0	0	1	1
ADNI3	168	Siemens	Prisma_fit	Accelerated	42	2.0	0	0	1	0	1
ADNI3	941	Siemens	Prisma_fit	Accelerated	55	2.0	1	0	0	0	1
ADNI4	389	Siemens	Prisma_fit	Accelerated	127	2.0	0	0	0	1	1
ADNI4	941	Siemens	Prisma_fit	Accelerated	127	2.0	0	0	0	1	1
ADNIGO/ADNI2	21	GE	DISCOVERY MR750	Accelerated	46	2.7	0	1	0	0	1
ADNIGO/ADNI2	27	GE	DISCOVERY MR750w	Accelerated	54	2.7	0	0	1	0	1
ADNIGO/ADNI2	37	Siemens	Prisma	Accelerated	127	2.0	0	0	1	0	1
ADNIGO/ADNI2	52	GE	DISCOVERY MR750	Accelerated	46	2.7	0	0	0	1	1
ADNIGO/ADNI2	52	GE	DISCOVERY MR750	Accelerated	68	2.7	0	0	1	0	1
ADNIGO/ADNI2	52	GE	DISCOVERY MR750	Accelerated	127	2.7	0	0	1	0	1
ADNIGO/ADNI2	57	GE	DISCOVERY MR750	Accelerated	54	2.7	1	0	0	0	1
ADNIGO/ADNI2	68	Siemens	Verio	Accelerated	55	3.0	1	0	0	0	1
ADNIGO/ADNI2	98	GE	DISCOVERY MR750	Accelerated	54	2.7	1	0	0	0	1
ADNIGO/ADNI2	99	GE	DISCOVERY MR750	Unaccelerated	54	5.0	1	0	0	0	1
ADNIGO/ADNI2	109	GE	Signa HDxt	Accelerated	46	2.7	0	0	0	1	1
ADNIGO/ADNI2	129	GE	SIGNA HDx	Accelerated	46	2.7	1	0	0	0	1
ADNIGO/ADNI2	135	GE	DISCOVERY MR750w	Accelerated	36	2.7	0	0	1	0	1

Note: Each row is one observed combination of phase, site, scanner manufacturer and model, acceleration category, final NIfTI-derived diffusion-volume count, and slice thickness. The 84 observed cells comprised 0 cells containing all four groups, 12 containing three groups, 34 containing two groups, and 38 containing one group, demonstrating incomplete crossing of diagnosis and acquisition factors.

Table S19: Multiplicity families used in the revised analyses.

Analysis family	Tests	Definition	Correction
Aggregated networks	16	Eight networks, separately for average- and modal-rank endpoints	BH-FDR within endpoint (8)
Tian structure omnibus	14	Seven structures, separately for average- and modal-rank endpoints	BH-FDR within endpoint (7)
Primary hippocampal pairwise	12	Six contrasts for each primary rank endpoint	BH-FDR across 12
Raw-value omnibus	2	Raw average and modal controllability	BH-FDR across 2
Density-adjusted raw omnibus	2	Density-adjusted raw average and modal controllability	BH-FDR across 2
Raw-value pairwise	12	Six contrasts for each raw endpoint	BH-FDR across 12
Hippocampal long axis	8	Four subdivisions, separately by endpoint	BH-FDR within endpoint (4)
Cognition associations	10	Two rank-based controllability endpoints by five cognitive outcomes, adjusted for age, sex, and education	BH-FDR across 10
Matched FreeSurfer cognition	15	Three imaging measures by five cognitive outcomes	BH-FDR across 15
Topology group contrasts	4	Four exploratory contrasts for hippocampal betweenness	BH-FDR across 4
Topology–controllability	4	Two unranked endpoints by strength and betweenness	BH-FDR across 4
Incremental cognitive models	5	Five cognitive outcomes	BH-FDR across 5
Biomarker sensitivity	16	Two endpoints within biomarker, window, and tested-effect strata	BH-FDR within each 2-test family
Low-coverage exclusion sensitivity	2	Average- and modal-rank group effects after excluding five review-trigger cases	BH-FDR across 2
Low-dimensional classification	24	Six feature sets in four tasks	Holm across 24
Paired classification comparisons	20	Five candidate-versus-volume comparisons in four tasks	Holm across 20

Note: BH-FDR, Benjamini–Hochberg false discovery rate correction; Holm, Holm familywise error rate correction.

Table S20: Acquisition-overlap diagnostics and design-balanced sensitivity analyses.

A. Omnibus results and balance diagnostics

Analysis	Endpoint or diagnostic	n or ESS	Statistic	p	p_{FDR}	Partial η^2
Strict covariate adjustment	Average rank	281	$F_{3,210} = 4.368$	0.0057	0.0093	0.059
	Modal rank	281	$F_{3,210} = 3.977$	0.0093	0.0093	0.054
Generalized overlap weighting	Overall effective sample size	45.2	Max pairwise SMD: 1.640 to 1.042	–	–	–
	Average rank	45.2	Wald $\chi^2_3 = 8.875$	0.0310	0.0310	0.098
	Modal rank	45.2	Wald $\chi^2_3 = 8.979$	0.0296	0.0310	0.099
Connected exact-cell common support	Average rank	218	$F_{3,164} = 5.663$	0.0014	0.0025	0.094
	Modal rank	218	$F_{3,164} = 5.112$	0.0025	0.0025	0.086
All-four-groups-per-cell restriction	Feasibility diagnostic	0	No exact cell contained all four groups	–	–	–

B. Covariate-adjusted estimates in the connected exact-cell common-support sample

Endpoint	Group	n	Adjusted mean	95% CI
Average rank	CN	93	345.73	325.44–366.01
	SMC	24	383.14	340.41–425.87
	MCI	66	321.88	297.12–346.63
	AD	35	286.03	255.84–316.23
Modal rank	CN	93	107.18	86.21–128.16
	SMC	24	69.41	25.86–112.95
	MCI	66	131.06	105.22–156.91
	AD	35	165.42	134.52–196.32

Note: The strict model included 12 scanner-model, three phase, 43 site, two acceleration, nine NIfTI-derived diffusion-volume-count, and four slice-thickness levels, together with age, sex, education, and two motion measures; the nominal 76-column full design had rank 71 with 210 residual degrees of freedom. Group-specific overlap-weighting effective sample sizes were 34.8 CN, 11.5 SMC, 6.7 MCI, and 20.9 AD. The weighted partial-η^2 bootstrap 95% CIs were 0.032–0.298 for average rank and 0.031–0.292 for modal rank. Residual balance remained inadequate, so weighted results are diagnostic irrespective of their nominal tests. The connected analysis retained 46 exact acquisition cells containing at least two groups and required direct shared-cell support for all six group pairs; permutations were restricted within cell. Direct SMC–MCI support comprised four cells with five SMC and six MCI participants. No exact cell contained all four groups. BH-FDR correction was applied across the two endpoints within each estimable analysis. SMD, standardized mean difference; ESS, effective sample size.

Table S21: Diffusion-protocol distribution and retrospective registration/label-placement QC.

Item	Result
Software	MRtrix3 3.0.6; FSL 6.0.7.18, including FLIRT
NIfTI-derived total-volume categories	22 (1), 31 (44), 36 (24), 42 (1), 46 (61), 54 (39), 55 (87), 68 (9), and 127 (15)
Observed b-value shell sets	$b = 0/1000$, $b = 0/1300$, and $b = 0/500/1000/2000$ s/mm^2
Observed b0-count categories	1, 4, 5, 6, 7, 8, and 13
Observed nonzero diffusion-weighted volume-count categories	16, 30, 32, 36, 41, 48, 60, and 114
Protocol strata: $b = 0/1000$	257 participants: b0/nonzero counts 6/16 (1), 1/30 (44), 4/32 (24), 6/36 (1), 5/41 (61), 6/48 (39), and 7/48 (87)
Protocol strata: $b = 0/1300$	9 participants: b0/nonzero count 8/60
Protocol strata: $b = 0/500/1000/2000$	15 participants: b0/nonzero count 13/114
Response/FOD commands	`dwi2response dhollander` and `dwi2fod msmt_csd` were applied to every dataset
Connectome scaling	`-scale_invnodevol`: $A_{ij} = 2\sum_{s \in S_{ij}} w_s/(N_i + N_j)$, where N_i and N_j are node-label voxel counts
Phase-encoding metadata	BIDS directions i (2), j (244), and $j-$ (35) in the retrospective input audit; all 15 127-volume inputs were PA; reverse phase-encoding pairs were not used for susceptibility correction
Retrospective quantitative QC completed	281/281 participants
Brain-mask Dice	Mean 0.917; 0/281 below 0.70
Hippocampal-label coverage in DWI brain mask	Mean 0.992; 5/281 below 0.90 (CN 1, SMC 0, MCI 2, AD 2)
Manual review of coverage-trigger cases	Acceptable T1–DWI registration and hippocampal-label placement confirmed in all five cases; no gross misregistration or erroneous hippocampal localization
Bilateral hippocampal labels present	281/281 participants
QC montage generation	Registration, 281/281; hippocampal placement, 280/281; the one montage-generation failure was display-only and retained complete quantitative QC
Existing connectome availability and numerical checks	281/281 matrices available; all passed dimension, finite-value, non-negativity, symmetry, and zero-diagonal checks
Additional exclusions during retrospective audit	Registration, 0; tractography/connectome availability, 0; QC thresholds, 0
Sensitivity excluding five coverage-trigger cases	$n = 276$ (CN 109, SMC 27, MCI 86, AD 54); average rank $F_{3,269} = 5.554$, $p_{\mathrm{FDR}} = 0.0017$, partial $\eta^2 = 0.058$; modal rank $F_{3,269} = 5.264$, $p_{\mathrm{FDR}} = 0.0017$, partial $\eta^2 = 0.055$

Note: Counts in parentheses are participants in the final NIfTI input categories. All 127-volume inputs were PA acquisitions and each contained 13 b0 and 114 nonzero diffusion-weighted volumes. Dice and coverage are gross-overlap checks and do not prove anatomical localization accuracy. QC thresholds were review triggers rather than post hoc exclusion rules. The two exclusion-sensitivity tests formed one BH-FDR family. Candidate-level failure counts preceding formation of the final 281-participant analytic cohort were not available in the archived processing records.

4 Supplementary Figures

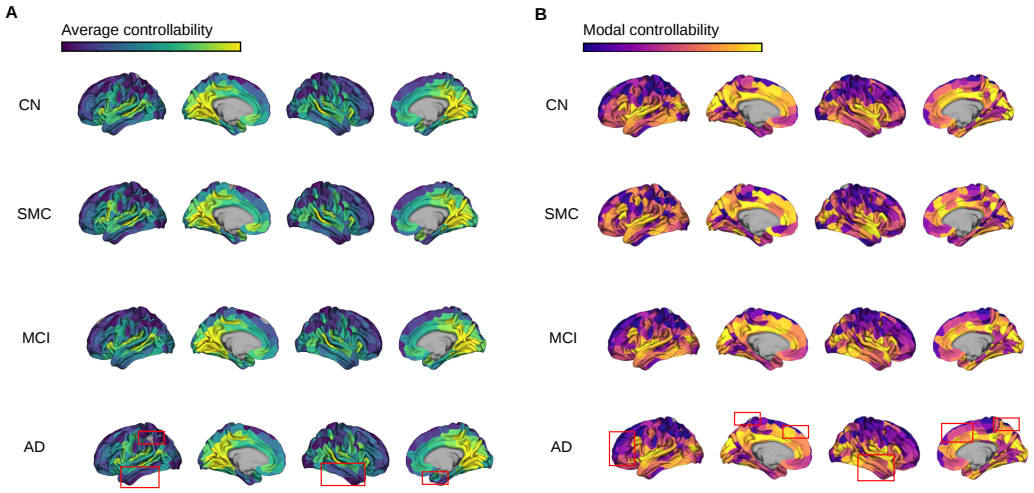


Figure S1: Descriptive cortical spatial distributions of average and modal controllability across clinically defined groups. (A) Average controllability and (B) modal controllability were averaged within each group and mapped to the Schaefer2018 cortical surface. A common color scale was used across CN, SMC, MCI, and AD within each metric to permit visual comparison. These maps are descriptive only; no parcel-level inference was performed, and visual differences should not be interpreted as statistically significant.

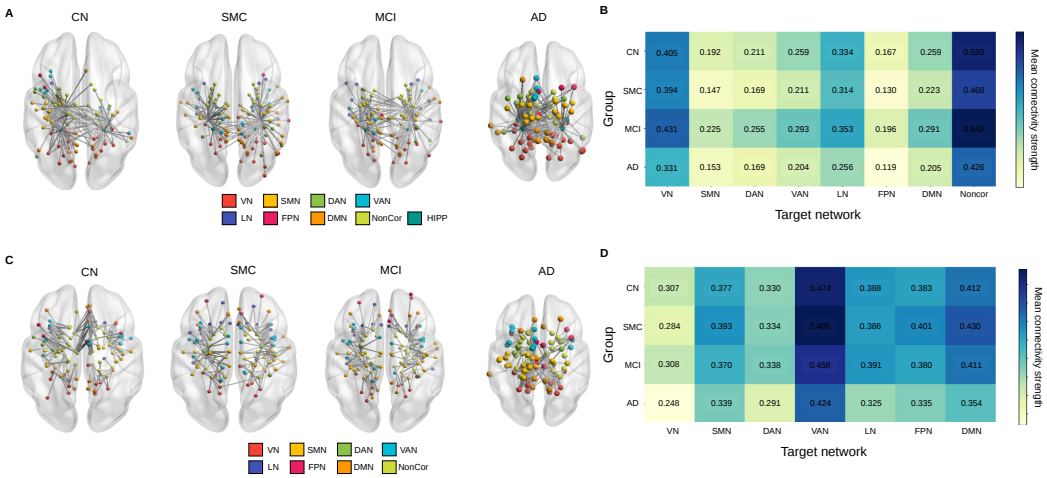


Figure S2: Descriptive structural-connectivity patterns of the hippocampus and Tian atlas-defined non-cortical structures across clinically defined groups. (A) Major hippocampal connections retained for visualization. (B) Mean connectivity between the hippocampus and the visual (VN), somatomotor (SMN), dorsal attention (DAN), ventral attention (VAN), limbic (LN), frontoparietal (FPN), default mode (DMN), and atlas-defined non-cortical (NonCor) target groups. (C) Major connections between Tian atlas-defined structures and cortical networks. (D) Mean connectivity between Tian atlas-defined structures and cortical networks. A common display threshold was used only to improve visualization and was not applied to inferential analyses; between-group visual differences should therefore be interpreted descriptively.

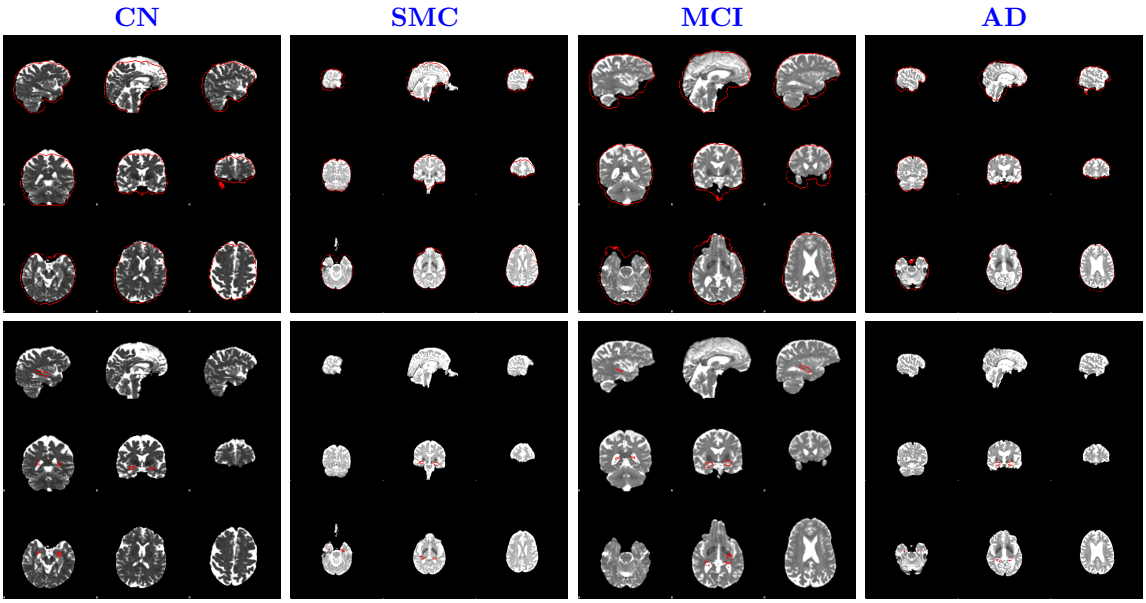


Figure S3: Representative retrospective registration and hippocampal-label-placement QC overlays. Columns show one CN, SMC, MCI, and AD participant selected near the cohort median brain-mask Dice value with complete hippocampal coverage. The top row shows the transformed T1-derived brain-mask contour over the mean b0 image; the bottom row shows transformed bilateral hippocampal-label contours over the mean b0 image. These examples illustrate the QC display, whereas quantitative registration and label-coverage metrics were computed for all 281 participants.

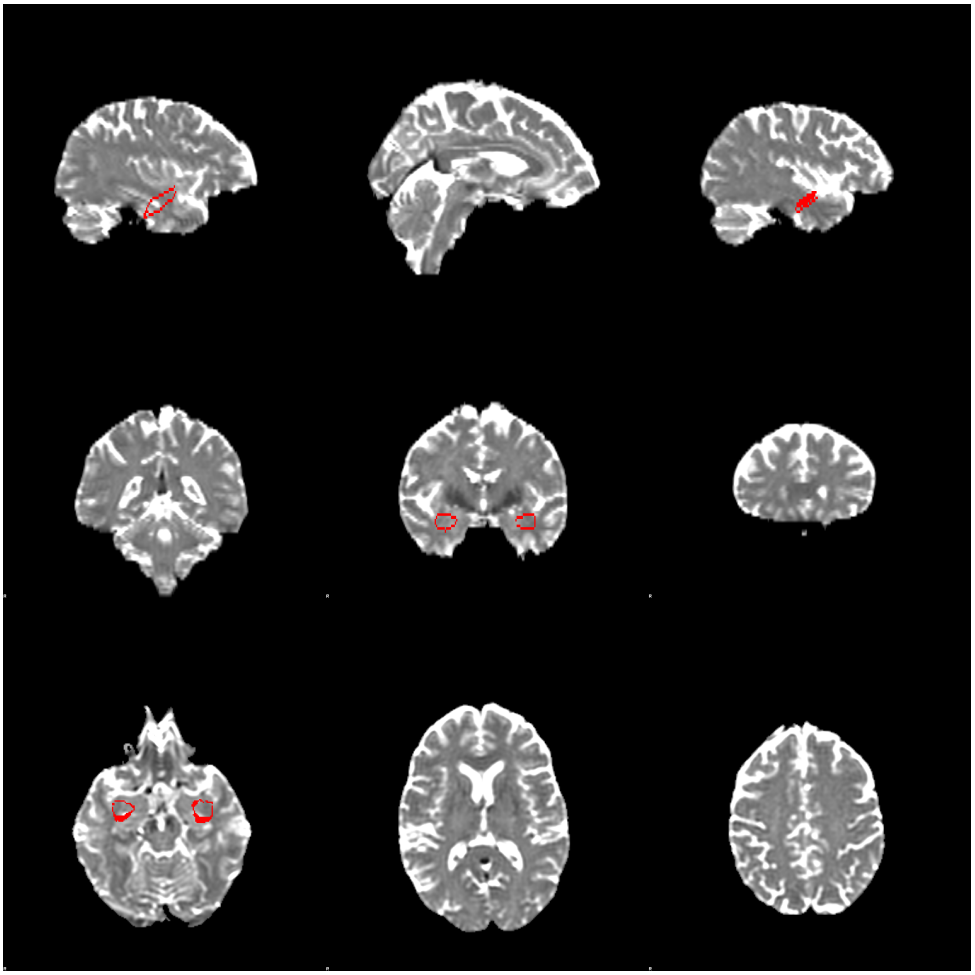


Figure S4: Example hippocampal-label overlay from a review-trigger case in the CN group. The transformed bilateral hippocampal-label coverage within the DWI brain mask was 0.894. Manual inspection confirmed acceptable T1–DWI registration and hippocampal-label placement without gross misregistration or erroneous hippocampal localization. The coverage threshold was used to trigger review rather than automatic exclusion; the primary group effects were retained after excluding all five coverage-trigger cases (Supplementary Table S21).